Project Documentation: GFC AI Financial Assistant (Prototype)

1. Methodology & Logic

The chatbot is architected as a Logic-Driven Retrieval System. Unlike prehistoric "if-else" hard-coded bots, this prototype leverages the Pandas library to treat financial data as a structured "Source of Truth."

• Data Ingestion: The system loads 10-K financial metrics (2022-2024) for Microsoft, Apple, and Tesla into a DataFrame.
• Intent Parsing: The bot uses string normalization and keyword matching to identify which company and which financial metric (Revenue, Net Income, or Margin) the user is interested in.
• Dynamic Calculation: It calculates Year-over-Year (YoY) growth and Profit Margins on the fly, ensuring that if the underlying data changes, the bot's insights remain accurate without rewriting code.

2. Predefined Queries Supported

The chatbot is optimized to respond to the following core inquiries:

• Total Revenue: "What is the total revenue for [Company]?"
• Net Income Trends: "How has net income changed for [Company]?"
• Profitability Analysis: "What is the profit margin for [Company]?"
• General Performance: "How is [Company] doing in 2024?"

3. Functionality & User Experience

To align with the Global Finance Corp brief, the bot avoids dense jargon. It provides:

• Readability: Large numbers are formatted with commas (e.g., $245,122M).
• Context: Instead of just a number, it explains if a metric has increased or decreased relative to the previous fiscal year.
• Interactivity: A continuous loop allows the user to ask multiple questions in a single session.

4. Limitations & Future Roadmap

• Static Scope: Currently limited to the three companies extracted from SEC EDGAR filings.
• Keyword Sensitivity: Requires specific keywords (Revenue, Income, Margin) to trigger data retrieval.
• Next Steps: Future iterations would implement RAG (Retrieval-Augmented Generation) to allow the bot to read full PDF reports and answer complex "Why" questions using a Large Language Model (LLM).